Buildings-Types of Buildings

Building

The Building is a kind of structure which is built with materials and including combination of structural elements such as foundation, column, beam, Slab, plinth, walls, floors, roofs, plumbing and building services etc.

For example, houses, factories, shopping malls, hospitals, etc. A building aims to give shelter along with security. Other purposes such as buildings serve several needs of society primarily as shelter from the weather, security, living space, privacy, to store belongings, supplied electricity, and to comfortably live and work.

Types of Buildings

1. Residential Buildings:

As the name suggests, the entire purpose of the residential building is significant for dwelling. It is considered in the category of residential buildings if more than half the area of the building is used for living. It is designed so that individuals and families stay in the house. Residential buildings, in general, have a living room, kitchen/cooking room, sleeping room, and bathing space.

Depending on the purpose it was constructed, there may be subcategories within the types of residential buildings such as houses, bungalows, apartments, duplex houses, hotels, dormitories, and so on.

2. Educational Buildings:

The types of educational buildings are purely built for the purpose of education or schooling. These building categories include schools, daycare, college, part-time teaching spaces, technical educational institutions, universities, and so on. The educational facilities can be both public and private in nature.

3. Institutional Buildings:

Institutional buildings generally fall under the category of civic buildings. This is because they are built to serve communities and ordinary people. The institutional buildings may be owned by both state/government and private. The common categories under types of institutional buildings include hospitals, museums, courts, post offices, prisons, libraries, community shelters, etc.

4. Assembly Buildings:

Assembly type of buildings is generally built for gatherings. These gatherings may be for the purpose of recreation, religion, or social uses/causes, and the common assembly buildings include hotels, restaurants, clubs, theatres, exhibitions, auditoriums, and so on. Different sections of people in society use the assembly buildings for different individual purposes.

5. Business Buildings:

The business buildings fall under the retail category. They can be an isolated structure or group of shops and facilities. Common business or retail building structures include shops, malls, realtors, boutiques, etc. These include both small and high rise building structures and may consist of just a single structure or group of establishments in one place.

6. Mercantile Buildings:

The mercantile buildings generally include wholesale or retail shops and structures for the everyday purposes of commoners. These different kinds of buildings are trading spaces, which include facilities such as general stores. Therefore, it is among an everyday use building to facilitate and help the citizens of the society.

7. Industrial Buildings:

The industrial buildings are otherwise popularly known as manufacturing structures. These include facilities that are a hub for the manufacturing, such as factories, warehouses, storage facilities, power plants, etc. In these structures, materials or products are assembled or processed to gain an end product.

8. Storage Buildings:

As the name suggests, the storage structures are built exclusively for the storage of goods and merchandise. These include cold storage, stables, sheds, godams, etc. these buildings are specifically constructed to make goods long-lasting with the right equipment and facilities in the establishments.

9. Hazardous Buildings:

The hazardous buildings are carefully tackled to only look into handling and storage, manufacturing of poisonous or explosive materials. These are specifically structured to build facilities that manage and control chemicals, toxic liquids, and explosives.

The hazardous buildings generally come with a heavy safety mechanism, and ordinary citizens in the society are not allowed freely to move under this property, given the safety precautions. The owner of these buildings is by a specific authority, both public and private.

10. Recreational Buildings:

The recreational buildings are where can engage and consume in several activities for pleasure or entertainment. People or groups come to these building spaces for their personal purpose of leisure. These are public spaces, which may be both government and private. The common buildings in recreation include a swimming pool, park, and fitness centers.

Building Materials

Some of the common construction materials include as follows

1.Bricks

2.Aggregates

3.Cement

4.Concrete

5.Wood

6.Steel

7.Metals

8.Glass

9.Polymers

Bricks

Brick is a type of construction material made of clay (Red clay) used to build walls, pavements and other elements in masonry construction.

Types of Bricks

1. Types of Bricks based on Strength

2. Types of Bricks based on Quality

1. Types of Bricks based on Strength : BIS (Bureau of Indian standard) bricks are classified according to their strength.

Sl.No	Class	Compressive strength in N/mm^2
1.	Class 35	35
2.	Class 30	30
3.	Class 25	25
4.	Class 20	20
5.	Class 17	17
6.	Class 15	15
7.	Class 12	12
8.	Class 10	10
9.	Class 7.5	7.5

2. Types of Bricks based on Quality:

1) First class bricks

1.	Compressive strength	10 N/mm^2
2.	Water absorption 24 hrs	15 to 17 %
3.	Colour	Deep red
4.	Sound	Metallic ring sound
5.	Texture	Uniform
6.	Surface	Smooth rectangular
7.	Size	(190 x 90 x 90) mm
8.	Moulding	Table moulded
9.	Burning	Kilns or Chimney
10.	Use	Brick wall, Reinforced brick work

2) Second class bricks

1.	Compressive strength	7 N/mm^2
2.	Water absorption 24 hrs	18 to 20 %
3.	Colour	Copper colour
4.	Sound	Metallic ring sound
5.	Texture	Uniform
6.	Surface	Smooth rectangular
7.	Size	(190 x 90 x 90) mm
8.	Moulding	Ground moulded
9.	Burning	Kilns or Chimney
10.	Use	Unimportant masonry work

3) Third class bricks

1.	Compressive strength	Less than 7 N/mm^2
2.	Water absorption 24 hrs	20 to 25%
3.	Colour	Red yellow coloured
4.	Sound	Dull sound
5.	Texture	Uniform
6.	Surface	Smooth rectangular
7.	Size	(190 x 90 x 90) mm
8.	Moulding	Ground moulded
9.	Burning	Kilns
10.	Use	Temporary masonry structure

4) Fourth class bricks

1.	Compressive strength	Less than 3.5 N/mm^2
2.	Water absorption 24 hrs	More than 30%
3.	Colour	reddish black colour
4.	Sound	Dull sound
5.	Texture	Non-Uniform
6.	Surface	Un-Smooth
7.	Size	(190 x 90 x 90) mm
8.	Moulding	Ground moulded
9.	Burning	Kilns
10	Use	Filling of ground area for floor

5) Special type of bricks

1.	Compressive strength	40 N/mm^2
2.	Water absorption 24 hrs	5 %
3.	Colour	Deep red
4.	Sound	Metallic ring sound
5.	Texture	Uniform
6.	Surface	Smooth rectangular
7.	Size	(190 x 90 x 90) mm
8.	Moulding	Table moulded
9.	Burning	Kilns
10.	Use	Important structural like Dam, Bridge

Characteristics of Good Bricks

1. Bricks should be uniform in colour, size and shape.
2. Standard size of brick should be maintained that is **190 x 90 x 90 mm**
3. Standard size of Frog should be maintained that is **100 x 40 x 10 mm**
4. They should be sound and compact.
5. They should be free from cracks, air bubbles, stone etc. with sharp and square edges.
6. Bricks should not absorb more than ($^1/_5$) of their own weight of water when immersed in water for 24 hours (**15% to 20%** of dry weight).
7. Weight of one burnt clay bricks is **3 kg.**
8. Unit weight of burnt clay bricks should be **1600 to 1900 kg/m^3.**
9. The compressive strength of bricks should be in range of **(7.5 to 35 MPa).**
10. The percentage of soluble salts should not exceed **2.5%** in brunt bricks.
11. Brick should not change in volume when wetted.
12. Bricks should neither over burnt nor under brunt.
13. The thermal conductivity of bricks should be low.
14. Bricks should be sound proof.
15. Bricks should be non-inflammable and incombustible.
16. Bricks should be free from lime pitting.

Manufacturing process of bricks

1. **Preparation of clay**
    a) Un-soiling of clay
    b) Digging of ground
    c) Cleaning of clay
    d) Weathering of clay
    e) Blending of ingredients
    f) Tempering of clay
2. **Moulding**
    a) Hand moulding (for small scale)
        i. Ground moulded bricks
        ii. Table-moulded bricks
    b) Machine moulding (for large scale)
        i. Plastic clay machines
        ii. Dry clay machines
3. **Drying**
4. **Burning**

Types of Defects in Bricks

1. Over burning of bricks
2. Under burning of bricks
3. Bloating of bricks
4. Black core
5. Efflorescence
6. Brick cracking
7. Brick spalling
8. Chuffs
9. Lime blowing
10. Spots
11. Lamination bricks
12. Defects in brick size
13. Defects in brick shape

Aggregates: Aggregate is a hard material made with the hard stratum of rocks by disintegration.

1) **Aggregate**s is a crushed stone which used in concrete.
2) Aggregate in concrete occupies around 70-80 percent of the volume of concrete.
3) **Aggregates** are one of the important and essential constitutes of concrete.
4) **Aggregate** provides the body to the concrete
5) Aggregate reduces shrinkage of concrete.
6) Aggregate makes concrete strong and durable.

Types of Aggregates

1) Types of aggregates based on size.
2) Types of aggregates based on Shape

Types of Aggregates According to Size

Coarse Aggregates

a) The aggregate particle which retains on 4.75 mm sieve is known as Coarse aggregate.
b) it occupies 45% to 60 % volume of total aggregates.

Fine aggregates

a) Aggregate which passed through 4.75 mm sieve know as Fine Aggregate.
b) It occupies 35% to 40 % volume of total aggregates.

Classification of Aggregates Based on Shape

Rounded Aggregate

a) Rounded aggregates are totally nature-made shape.
b) It is available in the form of seashore gravel.
c) It has voids (32–33%), hence offers more workability.
d) It perform well with a low water-cement ratio.
e) It has poor interlocking behaviour in concrete
f) It has weak bond strength.
g) It is not suitable for high-strength concrete.

Irregular Aggregates

a) Irregular aggregates are made by attrition.
b) It has partly round shape.
c) it has voids (35- 37%), hence offers lesser workability.
d) It has bond strength slightly good compared to rounded aggregate.
e) It is not suitable for high-strength concrete.

Angular Aggregates

a) Angular aggregate has well defined edges.
b) These are obtained by crushing the rocks.
c) it has voids (38-45%), hence offer less workability.
d) it gives high specific area.
e) It has good bond strength between the ingredients in concrete.

Flaky Aggregates

a) Flaky aggregate thickness is less compared to length and width.
b) Thickness is less than the 60% of its mean dimension.
c) It is not suitable for concrete making.

Elongated Aggregates

a) Elongated aggregate length is larger to compare to width and thickness.
b) Length of aggregate is greater than 180% of its mean dimension.
c) It is not suitable for concrete making.

Flaky and Elongated Aggregates

a) If length of aggregate is larger than its width and width is larger than its thickness then it is said to be flaky and elongated aggregates.
b) It is not suitable for concrete making.

Cement

1) Cement is a binder, a substance used for construction that **sets, hardens,** and **adheres** to other materials to bind them together.
2) Cement is used to bind sand and gravel together.
3) Cement mixed with water and fine aggregate produces **mortar** for masonry.
4) Cement mixed with water, sand and gravel, produces **concrete**.

Types of Cements used in the Construction industry

1. Ordinary Portland Cement (OPC)
2. Portland Pozzolana Cement (PPC)
3. Rapid Hardening Cement
4. Extra Rapid Hardening Cement
5. Quick Setting Cement
6. Sulfates Resisting Cement
7. Low Heat Cement
8. Blast Furnace Slag Cement
9. High Alumina Cement
10. White Cement
11. Coloured cement
12. Air Entraining Cement
13. Expansive cement
14. Hydrographic cement

Ordinary Portland Cement (OPC)

1) Ordinary Portland Cement is used worldwide for all general construction.
2) Annual global production of around 3.8 million cubic meters per year.
3) It is widely used for making Concrete, Mortar and Plaster.
4) Cement companies offer OPC in three different grades, namely **Grades 33, 43, and 53.**
5) Ordinary Portland cement is also used to manufacture grout, wall putty, solid concrete blocks and different types of cement also.

Portland Pozzolana Cement (PPC)

1) Portland pozzolana cement is prepared by grinding pozzolanic clinker with Portland cement.
2) PPC has a high chemical resistance on concrete.
3) It is widely used in construction such as:
    a) Marine structures
    b) Sewage works
    c) Bridges
    d) Piers
    e) Dams
    f) Mass concrete works.

Rapid Hardening Cement

1) Rapid hardening cement attains high strength in the early days.
2) The strength of rapid hardening cement at the three days is similar to 7 days strength of OPC with the same water-cement ratio.
3) it is used in concrete where formworks are removed at an early stage.
4) It is made when finely grounded C3S (Tricalcium Silicate) is displayed in OPC.
5) It is commonly used in rapid constructions like bridge pavement.

Extra Rapid Hardening Cement

1) As the name suggests, Extra rapid hardening cement gains strength quicker.
2) It is obtained by adding calcium chloride to rapid hardening cement.
3) It is about 25% faster than that of rapid hardening cement by one or two days.
4) It is widely used in cold weather concreting, to set the cement fast.

Quick Setting Cement

1) Quick setting cement which sets faster than OPC but the strength remains the same.
2) In this formula, the proportion of gypsum is reduced.
3) Quick setting cement is used for constructions that need a quick setting, like underwater structures and in cold and rainy weather conditions.

Sulfates Resisting Cement

1) This type of cement is manufactured to resist sulphate attack in concrete.
2) It has a lower percentage of C3A (Tricalcium Aluminate) and C4AF (Tetracalcium aluminoferrite).
3) It is used for constructions in contact with soil or groundwater having more than 0.2% or 0.3% g/l sulphate salts respectively.
4) It can also be used in concrete surfaces subjected to alternate wetting and drying like bridge piers, retaining walls and culverts.

Low Heat Cement

1) Low heat cement that is prepared by keeping the percentage of tricalcium aluminate below 6% and by increasing the proportion of C2S (Dicalcium silicate).
2) It is important to know that it is less reactive and the initial setting time is greater than OPC.
3) This low heat cement is used in mass concrete construction like gravity dams.

Blast Furnace Slag Cement

1) This type of cement is manufactured by grinding the clinker with about 60% slag.
2) It is similar to Portland cement.
3) It is used for constructions where economic considerations are important.

High Alumina Cement

1) High alumina cement is obtained by mixing calcining bauxite and lime with clinker during the manufacturing process of OPC.
2) To be considered high alumina cement, the total amount of alumina content should be at least 32%, and the ratio of the weight of alumina to lime should be kept between 0.85 to 1.30.
3) The most common uses are in constructions that are subject to high temperatures like a workshop, refractory, and foundries.

White Cement

1) This type of cement is manufactured by using raw materials that are free from iron and oxide.
2) White cement needs to have lime and clay in a higher proportion.
3) It is similar to OPC but it is more expensive.

Coloured cement

1) It is produced by mixing 5- 10% mineral pigments with ordinary cement.
2) They are widely used for decorative works on floors.

Air Entraining Cement

1) Air entraining cement is produced by adding air-entraining agents during the grinding of clinker. Such as
   - Resins,
   - Glues,
   - Sodium salts of sulphates, etc.
2) This type of cement is especially suited to improve the workability with a smaller water-cement ratio and to improve frost resistance of concrete.

Expansive Cement

1) Expansive cement expands slightly with time and does not shrink during and after the time of hardening.
2) This cement is mainly used for grouting anchor bolts and prestressed concrete ducts.

Hydrographic cement

1) Hydrographic cement is prepared by mixing **water-repelling chemicals** and has high workability and strength.
2) It has the property of repelling water and is unaffected during monsoon or rains.
3) Hydrophobic cement is mainly used for the construction of water structures such as
   - Dams,
   - Water tanks,
   - Spillways,
   - Water retaining structures, etc.

Materials used in Cement Manufacturing

1. Raw materials used for manufacturing of Portland cement are found naturally in the earth's crust.
2. It is made by the combination of
   - Calcareous materials (oxide of Calcium and Magnesium)
   - Argillaceous materials (oxide of Alumina and Iron)
   - Siliceous materials (oxides of Silica)
   - Gypsum.
3. **Calcareous Materials:**
   - Calcareous Materials are compounds of Calcium and Magnesium.
   - **Example:** Limestones and chalk.
4. **Argillaceous Materials:**
   - Argillaceous Materials are mainly Silica, Alumina, and Oxides of Iron.
   - **Example:** Clay and shale.
5. **Siliceous materials:**
   - Siliceous materials are compounds of oxides of Silica.
   - **Example:** fly ash, micro silica, chalk,

Concrete

Concrete is a building material it is defined as homogeneous mixture of water, fine aggregate (sand), coarse aggregate and cement.

Types of Concrete

1. Plain/ordinary concrete
2. Lightweight concrete
3. Precast concrete
4. Prestressed concrete
5. Self-compacting concrete
6. High density concrete
7. Reinforced concrete

Plain/ordinary concrete

This type of concrete is one of the most commonly-used, often for the construction of pavements and where buildings don't require very high tensile strength. The constituents are cement, sand and aggregate, mixed with water, typically in the ratio 1:2:4.

Lightweight concrete

Lightweight concrete, sometimes referred to as cellular concrete is a general term covering concrete solutions with reduced weights . which includes light weight aggregate concrete, no-fines concrete, aerated concrete, autoclaved aerated concrete as well as other similar products with varied properties.

Precast concrete

This is a form of concrete that is prepared, cast and cured off-site, usually in a controlled factory environment, using reusable moulds. Precast concrete elements can be joined to other elements to form a complete structure. They are typically used for structural components such as; wall panels, beams, columns, floors, staircases, pipes, tunnels, and so on.

Prestressed concrete

Prestressed concrete is a structural material that allows for predetermined, engineering stresses to be placed in members to counteract the stresses that will occur when they are subject to loading. It combines the high strength compressive properties of concrete with the high tensile strength of steel.

Self-compacting concrete

The introduction of self-compacting concrete (SCC) is regarded by some as one of the most important recent advancements in the concrete technology. It is a non-segregating concrete that can flow under its own weight, spread, fill formwork, and encapsulate reinforcements without the need for mechanical consolidation.

High density concrete

This type of heavyweight concrete has a greater density than other types and is manufactured using crushed rocks as coarse aggregate. As it provides good protection from x-rays and radiation, it is often used in nuclear power plants and other such buildings.

Reinforced concrete

Reinforced concrete (RC) is a versatile composite and one of the most widely used materials in modern construction. To increase its overall strength, steel rods, wires, mesh or cables can be embedded in concrete before it sets (or laid before the concrete is poured).